YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 4 SOLUTIONS

Statistics

Question 1

Data obtained from a group of people at a local health club are shown in the following table.

	Sex	Height (cm)	Weight (kg)	Fat (%)	Smoker
1	M	184	75.2	15.6	N
2	F	161	60.3	29.6	N
3	F	172	66.2	26.2	Y
4	M	189	92.6	27.2	N
5	M	179	76.6	19.2	N
6	M	172	83.5	8.2	N
7	M	184	68.9	9.1	N
8	F	165	63.7	22.3	N
9	F	169	62.4	24.6	N
10	F	158	51.1	18.3	Y
11	M	192	88.7	23.7	N
12	F	165	68.4	31.3	N
13	M	176	82.0	26.3	Y
14	F	170	55.2	20.8	N
15	F	161	59.8	25.1	N
16	M	182	73.5	19.2	N
17	F	167	53.9	19.9	Y
18	M	170	59.8	15.2	N
19	M	180	81.3	26.2	N
20	M	183	77.6	21.5	N
21	F	162	58.1	24.7	N
22	F	167	62.3	25.2	N
23	F	170	60.8	23.1	Y
24	M	186	89.5	11.6	N
25	F	159	50.7	20.6	N
26	F	164	59.7	24.7	N
27	M	177	68.9	18.2	Y
28	F	186	72.1	23.2	N
29	F	167	62.2	26.9	N
30	F	165	64.8	23.5	Y
31	M	187	86.2	12.6	N
32	F	159	63.2	30.6	N
33	M	169	73.2	11.8	N

Continued

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	Sex	Height (cm)	Weight (kg)	Fat (%)	Smoker
34	F	163	54.0	21.7	N
35	M	185	82.3	22.9	N
36	M	167	76.4	27.1	N
37	M	184	77.9	16.2	N
38	F	168	62.5	28.3	N
39	F	163	57.8	27.1	N
40	M	186	89.6	26.9	N

Consider the heights of the women at the health club.

a What type of data is this?

Numerical continuous

1 mark

b Find the mean, median and modal height, correct to one decimal place.

Mean = 165.8 cm; median = 165.0 cm and modes are 165.0 and 167.0.

3 marks

c Find the range and the standard deviation of the heights, correct to one decimal place.

Range = 28.0 cm; standard deviation = 6.1

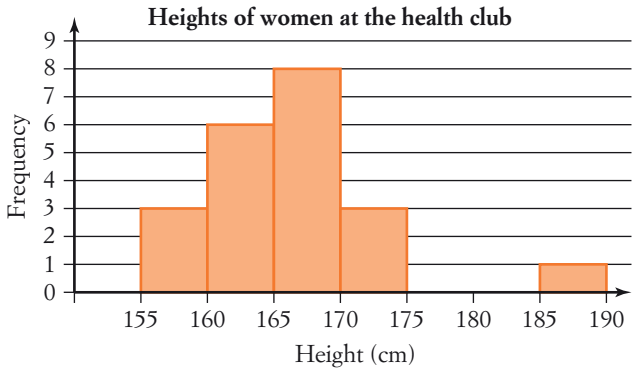
2 marks

d To display the data, create a frequency table with class intervals of 5 cm.

Class	Frequency
155 −<160	3
160 −<165	6
165 −<170	8
170 −<175	3
175 −<180	
180 −<185	
185 −<190	1

2 marks

e Draw a histogram to represent the data.



2 marks

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f Describe the shape of the histogram.

Roughly symmetrical with an outlier

1 mark

g Comment on the distribution of the heights of the women at this health club.

The height distribution is roughly symmetrical with an outlier. The centre is around 165 cm. As a result of the presence of the outlier, it is better to use the median. The large range (from 158.5 to 186.5) is mainly caused by the outlier of the height, 186.5. The next greatest height is 172 cm, which reduces the range from 28 to 13.5 cm. The standard deviation is large relative to the range, which also points to the presence of the outlier.

2 marks

Question 2

Now, consider the heights of the men at the health club.

a Find the mean, median and modal height, correct to one decimal place.

Mean = 180.6 cm; median = 183.0 cm; mode = 184.0 cm.

3 marks

b Find the range and the standard deviation of the heights, correct to one decimal place.

Range = 25.0 cm, standard deviation = 7.1 cm

2 marks

c Create a stem-and-leaf plot to represent the data.

Key: 16|7 = 167 cm

Stem	Leaf
16	7 9
17	0 2 6 7 9
18	0 2 3 4 4 4 5 6 6 7 9
19	2

2 marks

d Draw a pair of parallel boxplots to represent the two sets of data (both men and women), demonstrating and marking in the presence of any outliers.

Men: IQR = 186 − 176 = 10.

$Q_1 - 1.5 \times \text{IQR} \leq x \leq Q_3 + 1.5 \times \text{IQR}$

$176 - 1.5 \times 10 \leq x \leq 186 + 1.5 \times 10$

$161 \leq x \leq 201$

There are no outliers in the men's data.

Women: IQR = 168.5 − 161.5 = 7.

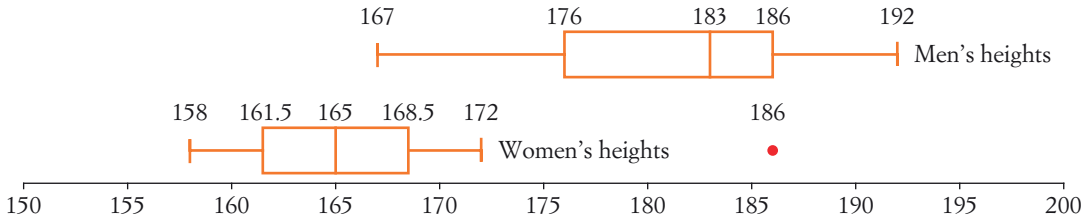
$Q_1 - 1.5 \times \text{IQR} \leq x \leq Q_3 + 1.5 \times \text{IQR}$

$161.5 - 1.5 \times 7 \leq x \leq 168.5 + 1.5 \times 7$

$151 \leq x \leq 179$

There is a single outlier (186 cm) in the women's data.

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4 marks

e Comment on the distribution of the heights of the women at this health club.

Generally speaking, men are taller than women (centre of 183 cm, versus 165 cm) and there is also greater variation in their heights (IQR of 10 cm versus 7 cm). The women have a roughly symmetrical distribution (with an outlier), whereas the men have a slightly negatively skewed distribution.

2 marks

Question 3

The owner of the health club is about to introduce a new program aimed at helping people to quit smoking, but first needs to assess the smoking status of its members. The owner wants to determine whether there is a relationship between gender and smoking status.

a Identify the explanatory and response variables in this potential relationship.

Explanatory variable = gender; response variable = smoking status.

1 mark

b Create a two-way table comparing gender and smoking status.

	Gender		
Smoking status	Male	Female	Total
Smoker	2	5	7
Non-smoker	17	16	33
Total	19	21	40

2 marks

c What percentage of:

i smokers are male

$\frac{2}{7} \times 100 = 28.57\%$

ii females are smokers

$\frac{5}{21} \times 100 = 23.81\%$

iii the group are smokers?

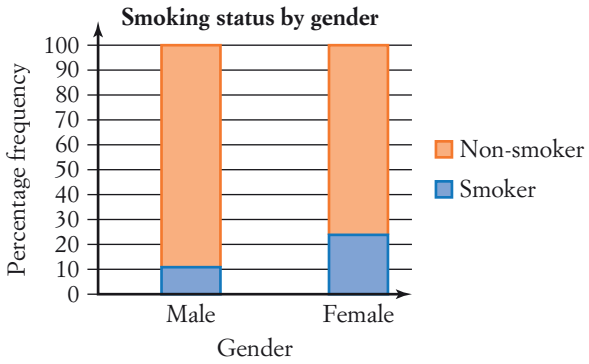
$\frac{7}{40} \times 100 = 17.5\%$

3 marks

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d Convert the two-way table into a percentage two-way table and hence draw a segmented column graph showing the smoking habits of each gender.

	Gender	
Smoking status	**Male**	**Female**
Smoker	$\frac{2}{19} \approx 10.5\%$	$\frac{5}{21} \approx 23.8\%$
Non-smoker	$\frac{17}{19} \approx 89.5\%$	$\frac{16}{21} \approx 76.2\%$
Total	100%	100%



3 marks

e Does there appear to be a relationship between gender and smoking status? Explain.

There may be a relationship between gender and smoking status, as approximately double the percentage of females in this sample smoke compared to the males.

1 mark

Question 4

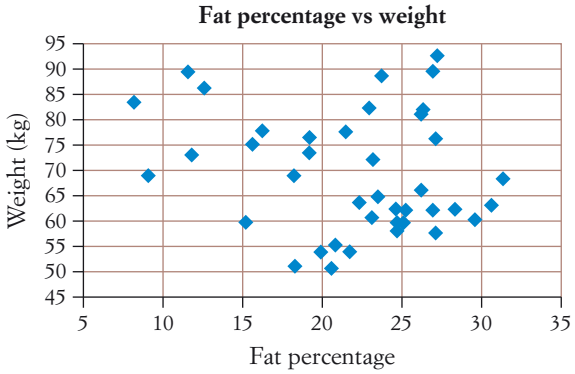
The owner of the health club wants to introduce a new weight-loss program and looks for a relationship between weight and fat percentage of participants.

a Identify the explanatory and response variables.

Explanatory variable = fat percentage; response variable = weight.

1 mark

b Draw a scatter plot showing the relationship between these two variables.



3 marks

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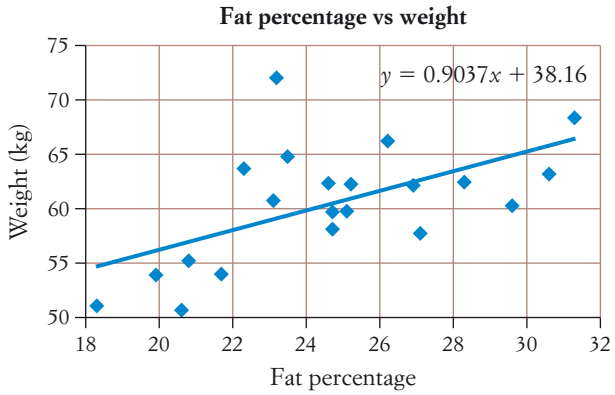
c Describe the scatter plot in terms of form, direction and strength. Does there appear to be a relationship between these two variables? Calculate Pearson's correlation coefficient to confirm.

The points are scattered randomly and there does not appear to be a relationship between the variables. Pearson's correlation coefficient $= -0.1968$, which confirms the lack of a relationship.

2 marks

Now, consider the points on the scatter plot that belong to the women.

d Redraw the scatter plot, using only the female data.



3 marks

e Describe the relationship that appears between these points in terms of form, direction and strength.

There appears to be a positive, moderately linear relationship between fat percentage and weight in the women.

1 mark

f Find Pearson's correlation coefficient, correct to four decimal places.

$r = 0.5822$

1 mark

g Find the equation of the least squares regression line and mark it in on your scatter plot.

Women's weight $= 0.9037 \times$ fat percentage $+ 38.1601$

2 marks

h What percentage of the variation in weight can be accounted for by the variation in body fat percentage?

$r^2 = 0.3389$.

Thus, 33.89% of the variation in the women's weights can be accounted for by the variation in their body fat percentage.

1 mark

i Predict the weight of a woman (correct to one decimal place) with a body fat percentage of:

i 25.8%

64.5 kg

ii 13.5%.

50.4 kg

2 marks

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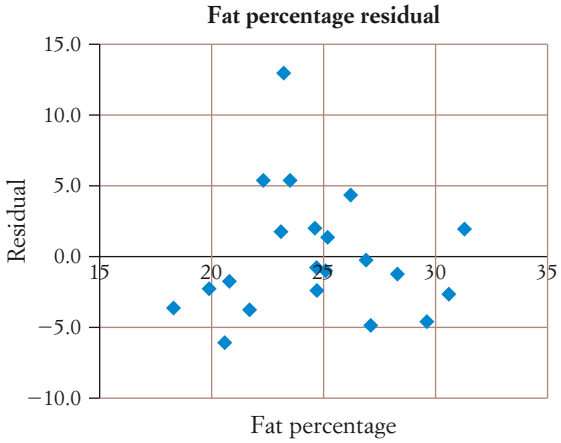
j Which prediction do you have most confidence in? Explain.

In the first prediction, as it is interpolation. The second prediction is extrapolation and, in particular, given that the relationship is on the weak side of moderate, making a prediction outside of the set is unlikely to be reliable.

1 mark

k Construct a residual plot and use this to determine whether the relationship is linear.

Fat (%)	Weight (kg)	Predicted weight	Residual
29.6	60.3	64.9	−4.6
26.2	66.2	61.8	4.4
22.3	63.7	58.3	5.4
24.6	62.4	60.4	2.0
18.3	51.1	54.7	−3.6
31.3	68.4	66.4	2.0
20.8	55.2	57.0	−1.8
25.1	59.8	60.8	−1.0
19.9	53.9	56.1	−2.2
24.7	58.1	60.5	−2.4
25.2	62.3	60.9	1.4
23.1	60.8	59.0	1.8
20.6	50.7	56.8	−6.1
24.7	59.7	60.5	−0.8
23.2	72.1	59.1	13.0
26.9	62.2	62.5	−0.3
23.5	64.8	59.4	5.4
30.6	63.2	65.8	−2.6
21.7	54.0	57.8	−3.8
28.3	62.5	63.7	−1.2
27.1	57.8	62.7	−4.9



The dots appear to be scattered randomly above and below the *x*-axis; hence, no pattern is evident. This suggests that the relationship is linear.

4 marks

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l Explain why, in order to see a pattern, each gender's points need to be considered separately.

Women have a higher fat percentage than men (for reproductive needs) and, so, when considered as a whole, there does not appear to be a pattern. A man and a woman with an equal fat percentage will likely have vastly different weights.

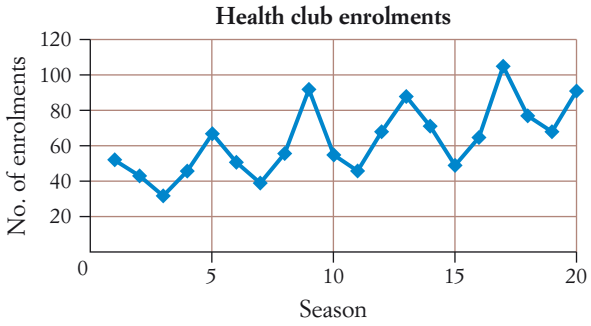
1 mark

Question 5

Enrolments at the health club over 5 years are recorded below.

Year	Jan	Apr	Jul	Oct
1	52	43	32	46
2	67	51	39	56
3	92	55	46	68
4	88	71	49	65
5	105	77	68	91

a Draw a time series plot using the table of data above.



3 marks

b Describe the trend shown in the plot.

Seasonal

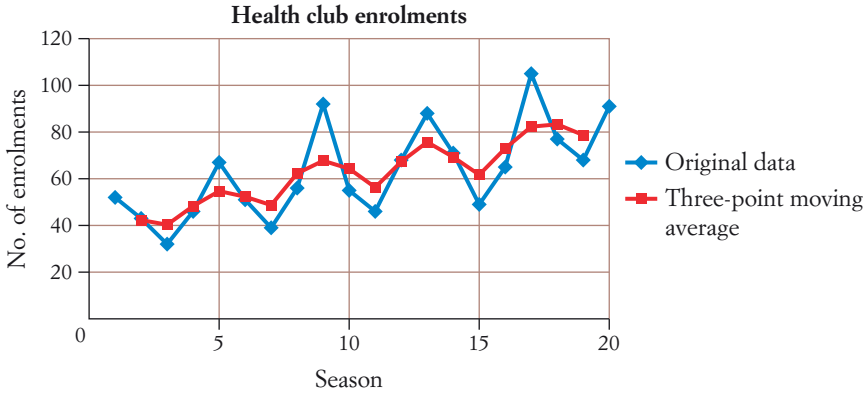
1 mark

c Perform a three-point moving average to smooth the data and plot the resulting data on the same graph as the original data.

Month	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr
No. enrolments	52	43	32	46	67	51	39	56	92	55
Moving average		42.3	40.3	48.3	54.7	52.3	48.7	62.3	67.7	64.3

Month	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct
No. enrolments	46	68	88	71	49	65	105	77	68	91
Moving average	56.3	67.3	75.7	69.3	61.7	73	82.3	83.3	78.7	

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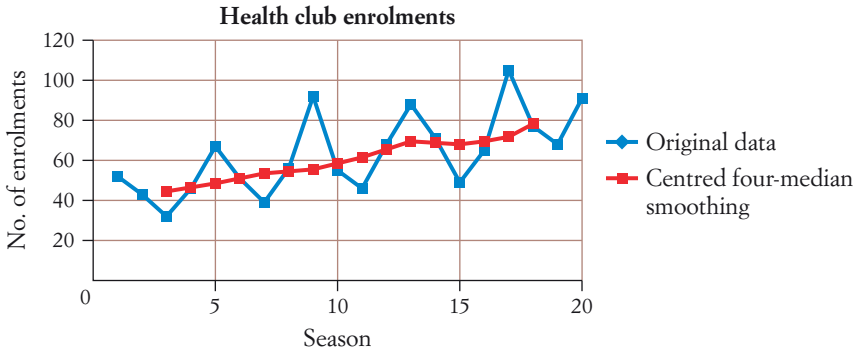


3 marks

d Perform centred four-point moving median smoothing to smooth the data and plot the resulting data on the same graph as the original data.

Month	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr
No. enrolments	52	43	32	46	67	51	39	56	92	55
Four-point median		44.5	44.5	48.5	48.5	53.5	53.5	55.5	55.5	61.5
			44.5	46.5	48.5	51	53.5	54.5	55.5	58.5

Month	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct
No. enrolments	46	68	88	71	49	65	105	77	68	91
Four-point median	61.5	69.5	69.5	68	68	71	72.5	84		
	61.5	65.5	69.5	68.75	68	69.5	71.75	78.25		



3 marks

e Which smoothing method provides the clearer view of the trend of the data? Explain.

The centred four-median smoothing method provides a clearer view of the trend of the data as it minimises the peaks and troughs of the seasons and shows a positive secular trend.

2 marks

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f Calculate the seasonal index (SI) for each quarter and deseasonalise the data.

Year	Jan	Apr	Jul	Oct	Yearly average
2010	52	43	32	46	43.25
2011	67	51	39	56	53.25
2012	92	55	46	68	65.25
2013	88	71	49	65	68.25
2014	105	77	68	91	85.25

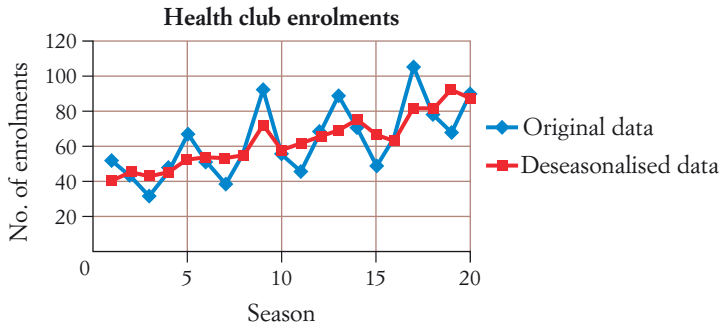
Year	Jan	Apr	Jul	Oct
1	$\frac{52}{43.25} = 1.2023$	$\frac{43}{43.25} = 0.9942$	$\frac{32}{43.25} = 0.7399$	$\frac{46}{43.25} = 1.0636$
2	$\frac{67}{53.25} = 1.2582$	$\frac{51}{53.25} = 0.9577$	$\frac{39}{53.25} = 0.7324$	$\frac{56}{53.25} = 1.0516$
3	$\frac{92}{65.25} = 1.4100$	$\frac{55}{65.25} = 0.8429$	$\frac{46}{65.25} = 0.7050$	$\frac{68}{65.25} = 1.0421$
4	$\frac{88}{68.25} = 1.2894$	$\frac{71}{68.25} = 1.0403$	$\frac{49}{68.25} = 0.7179$	$\frac{65}{68.25} = 0.9524$
5	$\frac{105}{85.25} = 1.2317$	$\frac{77}{85.25} = 0.9032$	$\frac{68}{85.25} = 0.7977$	$\frac{91}{85.25} = 1.0674$
Total	6.3915	4.7684	3.6929	5.1772
SI	$\frac{6.3915}{5} = 1.2783$	$\frac{4.7384}{5} = 0.9477$	$\frac{3.6929}{5} = 0.7386$	$\frac{5.1772}{5} = 1.0354$

Year	Jan	Apr	Jul	Oct
1	$\frac{52}{1.2783} = 40.68$	$\frac{43}{0.9477} = 45.37$	$\frac{32}{0.7386} = 43.33$	$\frac{46}{1.0354} = 44.43$
2	$\frac{67}{1.2783} = 52.41$	$\frac{51}{0.9477} = 53.82$	$\frac{39}{0.7386} = 52.80$	$\frac{56}{1.0354} = 54.08$
3	$\frac{92}{1.2783} = 71.97$	$\frac{55}{0.9477} = 58.04$	$\frac{46}{0.7386} = 62.28$	$\frac{68}{1.0354} = 65.67$
4	$\frac{88}{1.2783} = 68.84$	$\frac{71}{0.9477} = 74.92$	$\frac{49}{0.7386} = 66.34$	$\frac{65}{1.0354} = 62.78$
5	$\frac{105}{1.2783} = 82.14$	$\frac{77}{0.9477} = 81.25$	$\frac{68}{0.7386} = 92.07$	$\frac{91}{1.0354} = 87.89$

4 marks

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g Plot the deseasonalised data onto the same plot as the original data.



2 marks

h Find the least squares regression line for the deseasonalised data.

Deseasonalised data = 2.399 × season + 37.866

1 mark

i Use this regression line to find the predicted actual enrolment numbers for the summer of 2020.

Season = 41

Deseasonalised data = 2.399 × 41 + 37.866

= 136.225

Actual enrolment prediction = 136.225 × 1.2783 = 174.14

Predicted actual enrolment numbers for the summer of 2020 is 174 people.

2 marks

j If the health club can cater for 150 people at a time only, will they be able to cater for the predicted number of enrolments in the summer of 2020? Explain.

The health club will not be able cater for the predicted number of enrolments in the summer of 2020, as reseasonalising the deseasonalised prediction gives 174 enrolments, which is greater than the maximum of 150 enrolments for which the health club can cater.

1 mark

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